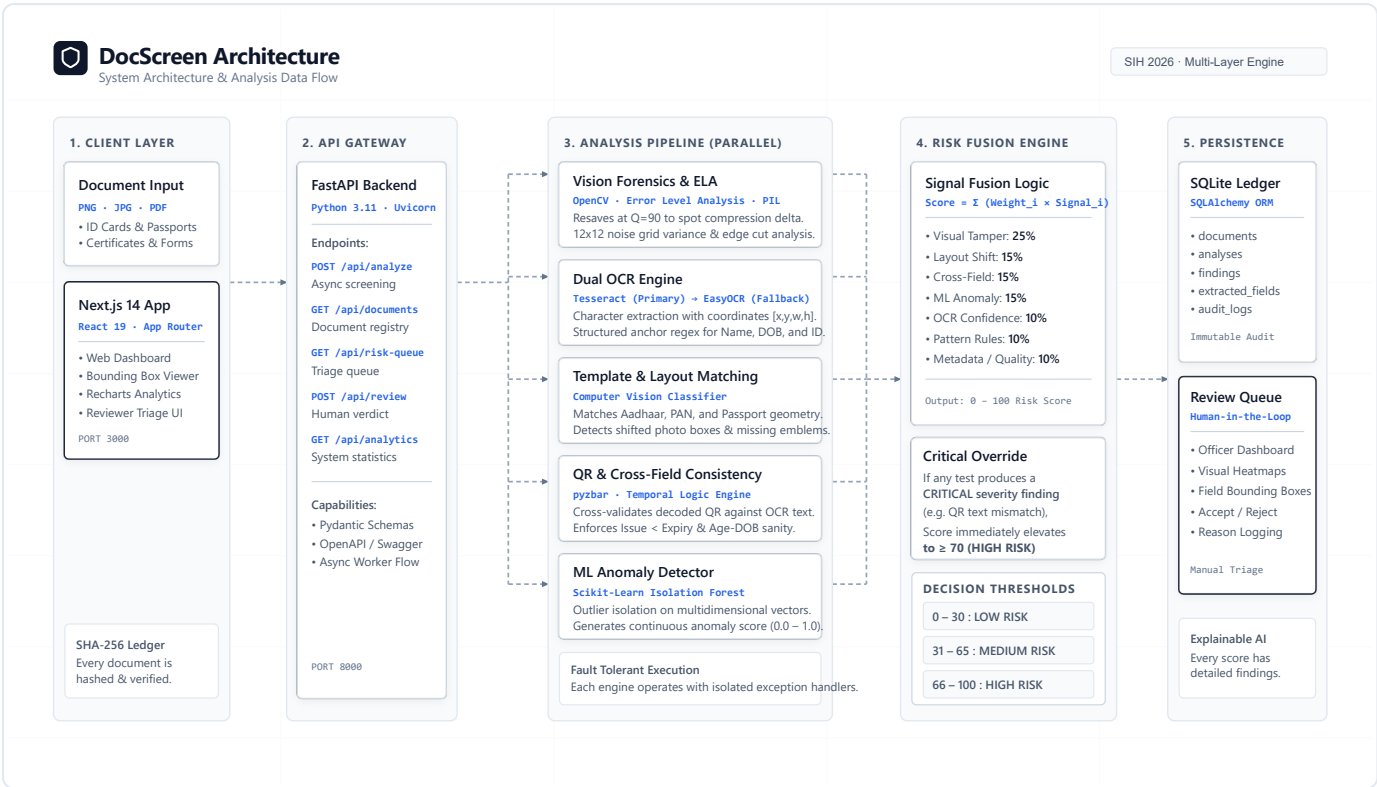


DocScreen — Architecture Brief

Multi-Layer AI Document Screening & Decision Fusion Pipeline

SIH Prototype Reference



1. WORKFLOW EXECUTION SUMMARY

1. Ingestion & Image Preprocessing

- SHA-256 Hashing:** Instant cryptographic duplicate detection.
- Quality Scoring:** Laplacian blur and luminosity histogram analysis.
- Contrast Normalization:** Adaptive CLAHE preprocessing.

3. Vision Forensics & Tamper Detection

- Error Level Analysis (ELA):** Identifies compression history anomalies from digital splices at Q=90.
- Noise Variance Grid:** 12x12 patch analysis to detect pasted elements.
- Edge Disparity:** Spots synthetic crop boundaries.

2. Dual OCR & Structured Extraction

- Dual-Engine Pipeline:** Tesseract primary with EasyOCR fallback.
- Spatial Extraction:** Word-level bounding boxes and confidence metrics.
- Regex Anchors:** Extraction of Name, DOB, and ID Numbers.

4. Consistency & ML Outlier Detection

- QR Cross-Validation:** Pyzbar decodes QR payload and verifies consistency against printed OCR text.
- Temporal Logic:** Validates Issue < Expiry and age-DOB checks.
- Isolation Forest:** Flags multidimensional outlier vectors.

Decision Fusion & Specifications

Weighted Risk Scoring, Override Logic, and Component Mapping

Technical Specifications

2. SIGNAL FUSION RISK ENGINE

The system synthesizes all detection outputs into a single index (0 – 100) using a weighted linear model:

SIGNAL	WEIGHT	TARGET EVALUATION
Visual Tamper	25%	ELA compression energy & noise variance inconsistencies
Template & Layout	15%	Shifted bounding boxes, missing official emblems, or margin modifications
Cross-Field Consistency	15%	QR code payload discrepancies vs printed OCR document text
ML Anomaly (Isolation Forest)	15%	Multidimensional outlier features compared to standard distributions
OCR Anomalies	10%	Character recognition confidence drops and distorted letterforms
Pattern Violations	10%	Checksum/regex failures (Aadhaar, PAN, Passport specifications)
Metadata & Quality	10%	EXIF software modification markers, blur, and glare penalties

3. DECISION CLASSIFICATION & OVERRIDES

LOW RISK (0 – 30) Standard verification; eligible for automated processing	MEDIUM RISK (31 – 65) Borderline confidence; queued for manual officer review	HIGH RISK (66 – 100) High likelihood of digital tampering or data contradiction
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Critical Finding Override

If any check identifies an unambiguous contradiction (e.g., **QR code payload disagrees with printed text** or **photo splice detected**), the overall score is automatically elevated to **≥ 70 (HIGH RISK)** regardless of other passing signals.

4. COMPONENT TECHNOLOGY STACK

LAYER	TECHNOLOGY	ROLE
Frontend	Next.js 14, React 19, Recharts	Web interface, bounding box viewer, analytics dashboard
Backend API	FastAPI, Uvicorn, Pydantic, Python 3.11	Asynchronous REST endpoints and analysis orchestration
Vision & Forensics	OpenCV, PIL, NumPy	Image enhancement, Error Level Analysis, noise variance mapping
OCR & Barcode	Tesseract OCR, EasyOCR, Pyzbar	Dual-engine optical character recognition and QR decoding
Machine Learning	Scikit-Learn (Isolation Forest)	Unsupervised multidimensional anomaly detection
Data Persistence	SQLite, SQLAlchemy ORM	Immutable audit logging and verification records